In [ ]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import MinMaxScaler
```

Collect Data

In [ ]:
```
df = pd.read_csv('/content/diabetes.csv')
```

In [ ]:
```
df.head()
```

Out[ ]:

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeF
0	6	148	72	35	0	33.6	
1	1	85	66	29	0	26.6	
2	8	183	64	0	0	23.3	
3	1	89	66	23	94	28.1	
4	0	137	40	35	168	43.1	

In [ ]:
```
df.shape
```

Out[ ]:
```
(768, 9)
```

In [ ]:
```
df.isnull().sum()
```

Out[ ]:

	0
Pregnancies	0
Glucose	0
BloodPressure	0
SkinThickness	0
Insulin	0
BMI	0
DiabetesPedigreeFunction	0
Age	0
Outcome	0

dtype: int64

Replace Null Values

In [ ]:
```
invalid_zero_cols = ['Glucose', 'BloodPressure', 'SkinThickness', 'Insulin', 'BM
df[invalid_zero_cols] = df[invalid_zero_cols].replace(0, np.nan)
```

In [ ]:
```
df.isnull().sum()
```

Out[ ]:

	0
Pregnancies	0
Glucose	5
BloodPressure	35
SkinThickness	227
Insulin	374
BMI	11
DiabetesPedigreeFunction	0
Age	0
Outcome	0

dtype: int64

In [ ]:
```
imputer = SimpleImputer()
df[invalid_zero_cols] = imputer.fit_transform(df[invalid_zero_cols])

df = df.drop_duplicates()
```

In [ ]:
```
df.isnull().sum()
```

Out[ ]:

	0
Pregnancies	0
Glucose	0
BloodPressure	0
SkinThickness	0
Insulin	0
BMI	0
DiabetesPedigreeFunction	0
Age	0
Outcome	0

dtype: int64

Normalize Data

In [ ]:
```
scaler = MinMaxScaler()
features = df.drop(columns=['Outcome'])
features_scaled = scaler.fit_transform(features)
```

In [ ]:
```
df = pd.concat([pd.DataFrame(features_scaled, columns=features.columns), df['Out
```

Validate Data

In [ ]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 768 entries, 0 to 767
Data columns (total 9 columns):
 #   Column                    Non-Null Count  Dtype  
---  ------                    --------------  -----  
 0   Pregnancies               768 non-null    float64
 1   Glucose                   768 non-null    float64
 2   BloodPressure             768 non-null    float64
 3   SkinThickness             768 non-null    float64
 4   Insulin                   768 non-null    float64
 5   BMI                       768 non-null    float64
 6   DiabetesPedigreeFunction  768 non-null    float64
 7   Age                       768 non-null    float64
 8   Outcome                   768 non-null    int64  
dtypes: float64(8), int64(1)
memory usage: 54.1 KB
```

In [ ]:
```
df.head()
```

Out[ ]:

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPe
0	0.352941	0.670968	0.489796	0.304348	0.170130	0.314928	
1	0.058824	0.264516	0.428571	0.239130	0.170130	0.171779	
2	0.470588	0.896774	0.408163	0.240798	0.170130	0.104294	
3	0.058824	0.290323	0.428571	0.173913	0.096154	0.202454	
4	0.000000	0.600000	0.163265	0.304348	0.185096	0.509202	

Store and Use

In [ ]:
```
df.to_csv("Cleaned_Diabetes_Dataset.csv", index=False)
```

Conclusion

In this lab, we learned how to build a complete healthcare data pipeline by collecting raw data, cleaning inconsistencies, integrating multiple sources, and transforming it into a machine-learning-ready format. This process ensures data quality and reliability for medical analytics.

Postlab

What is the risk of using unclean or unintegrated data in ML models?

- It can lead to **inaccurate predictions**, **biased models**, and **unreliable outcomes**.
- Common issues include **missing values**, **duplicate entries**, and **inconsistent formats**, which degrade model performance.

What are the main types of healthcare data sources?

- **Electronic Health Records (EHR)**
- **Lab Test Results**
- **Medical Imaging** (X-rays, MRIs, CT scans)
- **Wearable Device Data** (heart rate, steps, oxygen levels)
- **Public Health Databases** (government or open-access datasets)

What are the common challenges in collecting healthcare data?

- **Data Privacy and Compliance** (e.g., HIPAA, GDPR)
- **Inconsistent Data Formats** across hospitals and labs
- **Missing or Incomplete Data**
- **Unstructured Text Data** (like doctor's notes)
- **Integration Issues** when combining data from multiple sources

---

What are the common steps involved in cleaning healthcare data?

1. **Fix Missing Values** (e.g., fill or drop)
2. **Remove Duplicates**
3. **Correct Errors** (e.g., wrong entries, typos)
4. **Handle Outliers** (e.g., extreme test results)
5. **Standardize Units and Formats** (e.g., mg/dL vs. mmol/L)

In [ ]: